
Measurements & Instruments

A Solving Manual

IESE Manual 1 of 3

Typeset in L^AT_EX with circuitikz and pgfplots.

How to use this manual

This manual turns the measurement and instrument material into a solving reference. Each *type* of problem gets a **Protocol** (a numbered recipe), one fully **Worked Example** (the canonical one) and **Exercises** with solutions; theory sits in one paragraph at the point of use, with the derivation in a cross-referenced **appendix**.

Key Idea

A measurement is three things: a value, a unit, and an *uncertainty*. The number alone is never enough to make a yes/no decision. Most of Chapter 2 is about getting the uncertainty right — especially how it *propagates* when you combine measurements.

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Chapter 1

Measurements and the SI system

Before a single number is written down, two questions must already have answers: *in what unit?* and *traceable to what?* This chapter collects the metrological groundwork the subject opens with — the modern definition of the International System of Units (SI), how a measurement is organised, and how any reading taken in the lab is ultimately tied back to a universal standard.

Key Idea

A measurement is a *comparison*. Every value you report is implicitly the ratio of the measurand to a *unit*, so the unit must mean the same thing for everyone — which is what the SI guarantees — and you must be able to prove your instrument still agrees with that unit — which is what *traceability* does.

1.1 The 2019 redefinition: from artefacts to constants

For most of its history the SI rested on *artefacts*: physical objects whose properties *defined* a unit. The archetype was the international prototype of the kilogram (IPK), a platinum–iridium cylinder kept near Paris; the kilogram *was*, by definition, the mass of that object. Artefacts are fragile: they drift over decades, they can be damaged, and there is only one master — so the entire world’s mass scale depended on a single cylinder nobody could re-create.

On 20 May 2019 the SI was redefined so that every base unit is fixed by assigning an *exact* numerical value to a *fundamental constant of nature*. Nature is the same everywhere and at all times, so a unit defined this way is reproducible in any suitably equipped laboratory, with no master object to guard.

Note

The redefinition did not perceptibly change the *size* of any unit — a kilogram still weighs a kilogram. It changed the *definition*, trading a man-made object for an unchanging constant.

1.2 The seven base units

The SI is built on seven base units, each now pinned to a defining constant assigned an exact value. Every other unit is derived from these.

Note

The second and the speed of light come first for a reason: once the second is fixed by the caesium transition and c is fixed exactly, the metre is *defined* as the distance light travels in $1/c$ seconds. Several units chain off one another this way.

Table 1.1: The seven SI base units and their 2019 defining constants.

Quantity	Unit (symbol)	Fixed by the exact value of
Time	second (s)	$\Delta\nu_{\text{Cs}} = 9\,192\,631\,770\text{ Hz}$
Length	metre (m)	$c = 299\,792\,458\text{ m/s}$
Mass	kilogram (kg)	$h = 6.626\,070\,15 \times 10^{-34}\text{ J s}$
Electric current	ampere (A)	$e = 1.602\,176\,634 \times 10^{-19}\text{ C}$
Temperature	kelvin (K)	$k = 1.380\,649 \times 10^{-23}\text{ J/K}$
Amount of substance	mole (mol)	$N_A = 6.022\,140\,76 \times 10^{23}/\text{mol}$
Luminous intensity	candela (cd)	$K_{\text{cd}} = 683\text{ lm/W}$

1.3 Derived units

A derived unit is a product of powers of base units. Many keep an explicit form (area m^2 , velocity m/s); others earn a special name. The volt, for instance, is

$$\text{V} = \text{W/A} = \text{kg m}^2/(\text{A s}^3),$$

and the ohm, farad, hertz, newton, joule and watt are all shorthands for specific base-unit combinations. Tracking the dimensions through a formula (a *dimensional check*) is the cheapest sanity test there is: if the units on the two sides of an equation do not match, the equation is wrong.

1.4 The approach to a measurement

A measurement is not just "read the display". A defensible result follows a method, captured below as a reusable recipe.

Protocol 1.1 Take and report a measurement

1. **Define the measurand:** state exactly *what* quantity, under *what* conditions (temperature, loading, frequency).
2. **Choose the method** — direct or indirect (§1.6) — and an instrument whose range and resolution suit the expected value.
3. **Account for the instrument's influence:** a real meter *loads* the circuit; decide whether that perturbation matters (Chapter 5).
4. **Read value, unit and uncertainty** — never the bare number (Chapter 2).
5. **Propagate** the uncertainty if the result is computed from several readings (Chapter 2, Appendix A).
6. **Report** $x_0 \pm \Delta x$ with its unit, and the conditions.

1.5 Traceability

Metrological traceability is the property of a measurement result whereby it can be related to a stated reference through a documented, unbroken chain of calibrations, each contributing to the uncertainty. In practice the chain descends from the SI definition at the top to the instrument on your bench:

- **BIPM** and the defining constants — the SI itself;
- **National metrology institutes** (e.g. INRiM in Italy) realise the units and keep primary standards;
- **Accredited calibration laboratories** calibrate reference instruments against those primary standards;
- **Your working instrument**, calibrated against a reference, carries a certificate with its own uncertainty and a due date.

Watch Out

Traceability is broken the moment a link is undocumented or out of date. An expired calibration sticker means the chain to the SI is no longer proven — the reading may be fine, but you can no longer *defend* it.

1.6 Classification of measurement methods

Direct methods

compare the measurand against a reference of the *same kind*: a ruler against a length, a balance against a reference mass, a voltmeter that reads volts directly. The result is read in one step.

Indirect methods

obtain the measurand by *computing* it from other measured quantities through a known physical relation — e.g. a resistance from $R = V/I$ (measure V and I , then divide), or power from $P = V^2/R$. The catch is that every input's uncertainty propagates into the result (Chapter 2).

Worked Example 1.1 Classifying and tracing a resistance measurement

You read a resistor as $50.0\,\Omega$ on a digital multimeter (DMM) set to " Ω ", then also compute it from a separate V and I reading.

- The DMM reading is a **direct** method: the instrument compares the unknown against an internal reference and reports ohms in one step. Its result is **traceable** only if the DMM's calibration certificate is current.
- The $R = V/I$ route is an **indirect** method: two readings, each with its own uncertainty, combine. Because it is a ratio, the *relative* uncertainties add (Chapter 2):

$$\varepsilon_R = \varepsilon_V + \varepsilon_I.$$

Same quantity, two methods — the direct one is simpler, the indirect one exposes where the error comes from.

Exercise 1.1

Classify each as direct or indirect, and name the quantities whose uncertainty enters the result: (a) timing 10 pendulum swings with a stopwatch to get one period; (b) measuring a temperature with a calibrated thermometer; (c) finding the input resistance of an amplifier from the divider $V_M = V_S R_{in}/(R + R_{in})$.

Solution

(a) **Indirect**: you measure a total time t_{10} and divide, $T = t_{10}/10$; only t_{10} carries uncertainty (dividing by the exact integer 10 also divides the absolute uncertainty, which is

why you time ten swings). (b) **Direct:** the thermometer reads temperature in one step; traceable via its calibration. (c) **Indirect:** $R_{in} = V_M R / (V_S - V_M)$ is computed from V_M , V_S and the known R ; the difference $V_S - V_M$ in the denominator is the danger spot (Chapter 2).

Chapter 2

Measurements and uncertainty

Every measured value x comes with an uncertainty, written absolute ($x = x_0 \pm \Delta x$) or relative ($\varepsilon_x = \Delta x/x_0$, dimensionless). When a result is *computed* from several measured quantities, $y = f(x_1, \dots, x_n)$, the uncertainty propagates by a first-order expansion (Appendix A).

Protocol 2.1 Propagate uncertainty through $y = f(x_1, \dots, x_n)$

1. Write y as the combination of the measured quantities.
2. **Classify** the operation and apply the rule: *sum/difference* add *absolute* uncertainties $\Delta y = \sum |\partial f / \partial x_i| \Delta x_i$; *product/ratio* add *relative* uncertainties $\varepsilon_y = \sum \varepsilon_{x_i}$.
3. **Watch differences of close numbers:** $a - b$ with $a \approx b$ makes ε_y explode (catastrophic cancellation).
4. Report value, absolute and relative uncertainty.

The four elementary operations are applied to the *same* inputs — a useful calibration for your intuition before the surprising cases.

Worked Example 2.1 The four basic rules

Take $a = (15 \pm 0.5) \text{ cm}$ ($\varepsilon_a = 3.3\%$) and $b = (10 \pm 0.5) \text{ cm}$ ($\varepsilon_b = 5\%$).

- **Sum** $y = a + b = (25 \pm 1) \text{ cm}$: $\Delta y = \Delta a + \Delta b = 1 \text{ cm}$, $\varepsilon_y = 1/25 = 4\%$ — between ε_a and ε_b .
- **Difference** $y = a - b = (5 \pm 1) \text{ cm}$: the *same* $\Delta y = 1 \text{ cm}$, but now $\varepsilon_y = 1/5 = 20\%$ — worse than either input, because the value shrank while the uncertainty did not.
- **Product** ($a = b = (10 \pm 0.5) \text{ cm}$): $y = ab = (100 \pm 10) \text{ cm}^2$, $\varepsilon_y = \varepsilon_a + \varepsilon_b = 10\%$.
- **Ratio** (same inputs): $y = a/b = 1.0 \pm 0.1$, again $\varepsilon_y = 10\%$ — relative uncertainties add for both products and ratios.

Worked Example 2.2 Catastrophic cancellation, $y = 1/(a - b)$

$a = 10.00(2) \text{ cm}$, $b = 9.95(2) \text{ cm}$ (both $\varepsilon \approx 0.2\%$ — excellent). Yet $y_0 = 1/(a - b) = 1/0.05 = 20/\text{cm}$ and, since $\Delta y = \frac{\Delta a + \Delta b}{(a - b)^2} = \frac{0.04}{0.0025} = 16/\text{cm}$, we get $\varepsilon_y = 16/20 = 80\%$. Two superb measurements produced a useless result — because their difference is tiny.

Exercise 2.1

(a) $c = (22 \pm 0.5) \text{ cm}$, $d = (21 \pm 0.5) \text{ cm}$: find ε_z for $z = c - d$. (b) Measuring an input resistance via a known R and the divider $V_M = V_S \frac{R_{in}}{R + R_{in}}$: why is $R \approx R_{in}$ best?

Solution

(a) $z = 1$ cm, $\Delta z = \Delta c + \Delta d = 1$ cm, so $\varepsilon_z = 1/1 = 100\%$ — again a near-cancellation. (b) $R_{in} = \frac{V_M}{V_S - V_M} R$ has the difference $V_S - V_M$ in the denominator. If $R_{in} \gg R$ then $V_M \rightarrow V_S$ (denominator $\rightarrow 0$, error blows up); if $R_{in} \ll R$ then $V_M \rightarrow 0$ (ε_{V_M} blows up). The sweet spot is $R \approx R_{in}$.

Exercise 2.2 Powers count multiple times

A resistor's dissipation is computed as $P = V^2/R$ from $V = (10.0 \pm 0.2)$ V and $R = (50 \pm 0.5) \Omega$. (a) Find P , ΔP and ε_P . (b) Which measurement should you improve first, and why?

Solution

(a) V^2 is a product of V with itself, so ε_V counts *twice*: $\varepsilon_P = 2\varepsilon_V + \varepsilon_R = 2 \cdot 2\% + 1\% = 5\%$. With $P_0 = 100/50 = 2.00$ W this gives $P = (2.00 \pm 0.10)$ W. (b) The voltage: a power $y = x^n$ multiplies the relative uncertainty by n , so the 2% of V contributes 4% of the 5% total, while R contributes only 1%.

Chapter 3

The analog oscilloscope

In a CRT scope an electron beam is deflected horizontally by V_x and vertically by V_y through the deflection coefficients K_x, K_y (in V/div): the spot moves $x_{ndiv} = V_x/K_x$ and $y_{ndiv} = V_y/K_y$ divisions. The horizontal axis is normally driven by an internal sawtooth *time base*.

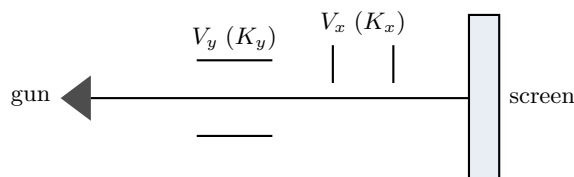


Figure 3.1: Cathode-ray tube: V_y and V_x deflect the beam vertically and horizontally; $x_{ndiv} = V_x/K_x$, $y_{ndiv} = V_y/K_y$.

3.1 Placing the beam and reading a trace

Protocol 3.1 Place the spot / interpret a trace

1. **Static spot:** $x_{ndiv} = V_x/K_x$, $y_{ndiv} = V_y/K_y$ from the screen centre.
2. **Time-varying** V_x, V_y : the spot traces a path; persistence and the eye merge it into a line above ~ 50 Hz.
3. **Stable image** needs (i) a periodic vertical signal and (ii) a time base *synchronised* by the trigger (level + edge), so each sweep redraws the same picture.

Worked Example 3.1 Time base + sinusoid (Example 9)

Time base: sawtooth, $K_x = 1$ V/div, $N_{xdiv} = 10$, so amplitude $A = K_x N_{xdiv}/2 = 5$ V, period $T = 100$ ms. Each horizontal division is $K_T = T/N_{xdiv} = 10$ ms/div. A sinusoid on V_y with $T_S = 50$ ms shows $T/T_S = 2$ full periods, and because the time base period is an integer multiple of T_S the picture is *stable*. Halving the scale (raise the time-base frequency to 200 Hz) shows one period.

Exercise 3.1

Sketch the spot for (a) $V_x = -2$ V, $V_y = 1.5$ V with $K_x = 0.5$, $K_y = 1$ V/div; (b) a square-wave V_y between 0 and 1 V, $K_y = 0.5$ V/div, $V_x = 0$. (c) Why does a time base period that is *not* an integer multiple of T_S give an unstable image (Example 10)?

Solution

(a) $x = -2/0.5 = -4$ div, $y = 1.5/1 = 1.5$ div: top-left quadrant. (b) The spot alternates between centre and 2 div up; above ~ 50 Hz both appear at once. (c) Each sweep starts at a different phase of V_y , so successive traces land in different places and the image drifts — fixed by a trigger or single-shot capture.

3.2 Choosing the scales

Protocol 3.2 Pick vertical and horizontal scales

1. **Vertical** (8 divisions): choose the smallest standard (1–2–5 sequence) $K_y \geq (V_{max} - V_{min})/8$ — fills the screen, minimising reading error.
2. **Horizontal** (12 divisions): to show N periods choose the smallest standard $K_x \geq NT/12$.
3. Set the trigger level inside the signal range, on the steeper edge.

Worked Example 3.2 Scale selection

A signal $V_{max} = 3$ V, $V_{min} = -3$ V: $(V_{max} - V_{min})/8 = 6/8 = 0.75$ V/div, so choose 1 V/div (signal fills 6 of 8 divisions). To show $N = 4$ periods of a 15 kHz signal ($T = 66.7$ μ s): $NT/12 = 22.2$ μ s/div, so choose 50 μ s/div.

Exercise 3.2 Scale selection

A signal swings between $V_{min} = -0.2$ V and $V_{max} = 1.0$ V at $f = 2$ kHz, and you want at least $N = 3$ periods on screen. (a) Pick the vertical scale. (b) Pick the horizontal scale and count the periods actually displayed. (c) What goes wrong with 100 μ s/div?

Solution

(a) $(V_{max} - V_{min})/8 = 1.2/8 = 0.15$ V/div: the next standard value up is 0.2 V/div (the signal then spans 6 of 8 divisions). (b) $T = 1/f = 500$ μ s, so $NT/12 = 125$ μ s/div: choose 200 μ s/div. The screen spans $12 \times 200 = 2.4$ ms, i.e. $2.4/0.5 = 4.8$ periods. (c) The screen would span only 1.2 ms = 2.4 periods, fewer than the 3 requested — 100 μ s/div is *below* the $NT/12$ minimum, not above it.

3.3 Reading uncertainty

A reading is a product, $V = K \cdot L$ (scale times divisions), so its relative uncertainty is the *sum* of the scale's calibration error and the reading error (Protocol of Chapter 2).

Protocol 3.3 Uncertainty of a scope reading

1. Express the reading as $V = K \cdot L$: scale (V/div or s/div) times length on screen (divisions).
2. Take the scale's calibration error ε_K from the datasheet (typically 1–3 %).
3. Estimate the reading resolution δL (typically 0.1–0.2 div, set by the graticule and

trace thickness); then $\varepsilon_L = \delta L/L$.

4. Add them: $\varepsilon_V = \varepsilon_K + \varepsilon_L$; report $V \pm \varepsilon_V V$.

Key Idea

δL is roughly *constant*, so the reading error $\varepsilon_L = \delta L/L$ shrinks as the trace gets bigger: **fill the screen**. The same trick works in time — measure the length of *several* periods and divide.

Worked Example 3.3 Peak-to-peak voltage

$K_y = 0.5 \text{ V/div}$ with $\varepsilon_{K_y} = 2\%$; the trace spans $L = 4 \text{ div}$ read to $\delta L = 0.2 \text{ div}$ ($\varepsilon_L = 0.2/4 = 5\%$). Then $V_{pp} = K_y L = 2 \text{ V}$, $\varepsilon_{V_{pp}} = \varepsilon_{K_y} + \varepsilon_L = 7\%$, $\delta V_{pp} = 0.14 \text{ V}$: $V_{pp} = 2.00(14) \text{ V}$.

Exercise 3.3 Period and frequency

With $K_T = 10 \mu\text{s/div}$ ($\varepsilon_{K_T} = 2\%$) one period of a signal spans $L = (5.0 \pm 0.1) \text{ div}$. (a) Find T and $f = 1/T$ with their uncertainties. (b) On $20 \mu\text{s/div}$, four periods span $(10.0 \pm 0.1) \text{ div}$ — redo the estimate and explain the improvement.

Solution

(a) $T = K_T L = 50 \mu\text{s}$, $\varepsilon_T = \varepsilon_{K_T} + \varepsilon_L = 2\% + 0.1/5.0 = 4\%$, so $T = (50 \pm 2) \mu\text{s}$. The ratio rule gives $\varepsilon_f = \varepsilon_T$ (the numerator 1 is exact): $f = (20.0 \pm 0.8) \text{ kHz}$. (b) $T = K_T L/4 = 50 \mu\text{s}$ again, but $\varepsilon_L = 0.1/10 = 1\%$ so $\varepsilon_T = 3\%$. Same δL , longer L : spreading the measurement over several periods divides the reading error by the period count.

Chapter 4

The digital oscilloscope

A Digital Sampling Oscilloscope samples the input with an ADC (N bits $\rightarrow \pm 0.5$ LSB quantisation) and reconstructs the trace from the samples. Two limits dominate: *aliasing* (sample too slowly) and the instrument's own *bandwidth* (which slows fast edges).

4.1 Sampling, Nyquist and aliasing

Protocol 4.1 Avoid aliasing; reconstruct cleanly

1. **Nyquist:** sample at $f_S > 2f_B$ (f_B = highest signal frequency); otherwise the trace aliases to a *wrong* signal with no warning.
2. **Points per period** for a faithful look: ~ 20 – 25 with no interpolator, ~ 10 with a linear one, 3–4 with a (sinc) one.
3. **Rule of thumb:** sample at $\geq 10\times$ the signal frequency.

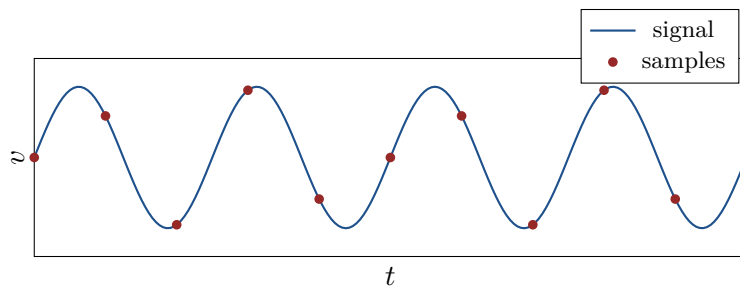


Figure 4.1: Sampling a sinusoid: enough samples per period let an interpolator recover the waveform; too few and the displayed trace is wrong (aliased).

Key Idea

A sine above the Nyquist limit does not vanish — it **folds back**: the samples of a sine at f_{in} are identical to those of a sine at

$$f_a = |f_{in} - kf_S|, \quad k \text{ the integer that puts } f_a \text{ in } [0, f_S/2].$$

The interpolator then draws a perfectly clean, perfectly *wrong* sine — with no clue of an error.

Worked Example 4.1 Predicting the aliased trace

A DSO samples at $f_S = 1$ MSa/s and the input is a sine at $f_{in} = 800$ kHz. Nyquist would need $f_S > 2f_{in} = 1.6$ MSa/s — violated. With $k = 1$, $f_a = |800 - 1000| = 200$ kHz: the

screen shows a clean 200 kHz sine of the *correct amplitude*, so nothing looks broken. To display the real 800 kHz signal acceptably with the default linear interpolator (~ 10 points per period) you need $f_S \approx 10f_{in} = 8 \text{ MSa/s}$.

Exercise 4.1 Alias frequencies

A DSO samples at $f_S = 50 \text{ MSa/s}$. (a) What does it display for a 49 MHz sine? (b) And for a 51 MHz sine — can you tell the two inputs apart? (c) What minimum f_S would you pick to view a 2 MHz sine with a linear interpolator?

Solution

(a) $k = 1$: $f_a = |49 - 50| = 1 \text{ MHz}$ — a clean, stable, wrong trace. (b) Also $f_a = |51 - 50| = 1 \text{ MHz}$: the two inputs produce *identical* samples, so they cannot be distinguished — aliasing destroys information. (c) Rule of thumb, 10 points per period: $f_S \geq 10 \times 2 = 20 \text{ MSa/s}$. (The theoretical Nyquist minimum 4 MSa/s is lossless only with the unfeasible ideal sinc interpolator.)

4.2 Probe loading and bandwidth

The scope input ($R_i \approx 1 \text{ M}\Omega$ in parallel with C_i) plus cable capacitance forms a low-pass that loads the source. A $\times 10$ **compensated probe** ($R_s C_s = R_i C_p$) attenuates by 10 but multiplies the bandwidth by 10 (Appendix B.1). The bandwidth B itself slows a step: the displayed rise time is $T_B = 0.35/B$, and a real edge T_{Rise} combines as $T_{meas} = \sqrt{T_B^2 + T_{Rise}^2}$.

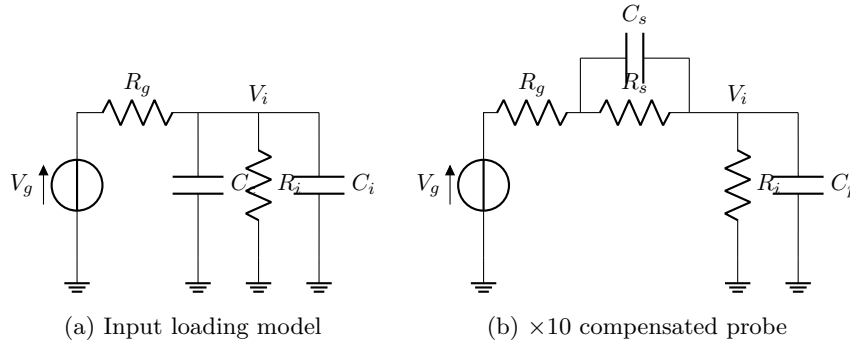


Figure 4.2: Cable + input capacitance load the source (a); a compensated probe with $R_s C_s = R_i C_p$ flattens the response and raises the usable bandwidth (b).

Protocol 4.2 Rise time and its uncertainty

1. Scope rise time $T_B = 0.35/B$.
2. Measured rise time $T_{meas} = N_{div} K_x$ (propagate $\varepsilon_{T_{meas}} = \varepsilon_{N_{div}} + \varepsilon_{K_x}$).
3. True edge $T_{Rise} = \sqrt{T_{meas}^2 - T_B^2}$; propagate to get $\varepsilon_{T_{Rise}}$.

Worked Example 4.2 Rise-time measurement (Example 1)

$B = 100 \text{ MHz} \pm 10\% \Rightarrow T_B = 0.35/B = 3.5 \text{ ns}$. Set $K_x = 10 \text{ ns/div}$ ($\varepsilon_{K_x} = 2\%$); read $N_{div} = (4.5 \pm 0.2) \text{ div}$ ($\varepsilon_{N_{div}} = 4.4\%$), so $T_{meas} = 45 \text{ ns}$, $\varepsilon_{T_{meas}} = 6.4\%$. Then $T_{Rise} = \sqrt{45^2 - 3.5^2} = 44.9 \text{ ns}$, and propagating gives $\varepsilon_{T_{Rise}} \approx 6.5\%$.

Exercise 4.2

Repeat with $K_x = 1 \text{ ns/div}$, same N_{div} in divisions (so $T_{meas} = 4.5 \text{ ns}$). Find T_{Rise} and comment on the uncertainty.

Solution

$T_{Rise} = \sqrt{4.5^2 - 3.5^2} = \sqrt{8} = 2.83 \text{ ns}$, but now T_B and T_{meas} are close, so the subtraction under the root is a near-cancellation: $\varepsilon_{T_{Rise}} \approx 32\%$. Measuring an edge near the scope's own limit is inaccurate — use a faster scope.

Chapter 5

The digital multimeter

5.1 Loading error and 2- vs 4-wire resistance

Connecting a meter changes the circuit. For voltage, the finite input resistance R_i forms a divider with the source resistance R_g : $V_m = V_g \frac{R_i}{R_i + R_g}$ (keep $R_i \gg R_g$). For resistance, the meter forces a known current and reads the voltage; in *2-wire* mode the lead resistance R_c adds ($R_{meas} = R_x + 2R_c$), while *4-wire* (Kelvin) sensing carries no current in the sense leads, so it cancels.

Protocol 5.1 Resistance measurement: choose 2- or 4-wire

1. Meter forces I_S , reads V_{meas} , reports $R = V_{meas}/I_S$.
2. **2-wire**: $R_{meas} = R_x + 2R_c$ — error grows for small R_x .
3. **4-wire**: sense leads draw no current, so their R_c drops no voltage; use it whenever R_x is small (down to Ω and below).

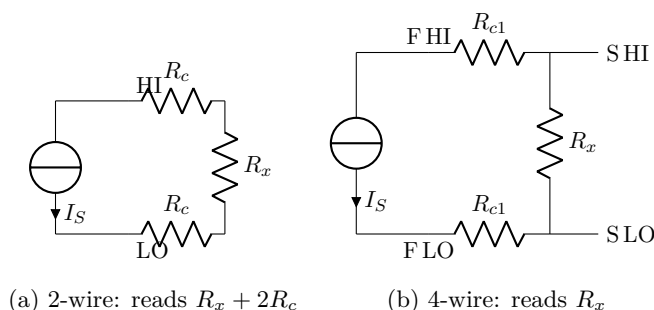


Figure 5.1: Resistance measurement. In 4-wire sensing the voltage is tapped right at R_x by leads that carry no current, so the lead resistances R_{c1} drop no voltage.

Worked Example 5.1 How big is the 2-wire error?

Measure $R_x \approx 1\ \Omega$ with ordinary leads, $R_c = 0.1\ \Omega$ each. The meter injects $I_S = 100\ \text{mA}$ (chosen so the drop is a few hundred mV). *2-wire*: $V_{meas} = I_S(R_x + 2R_c) = 0.1 \times 1.2 = 120\ \text{mV}$, so it reports $R_{meas} = 1.20\ \Omega$ — a +20% error. *4-wire*: the sense leads carry no current, so $V_{meas} = I_S R_x = 100\ \text{mV}$ and the meter reports $1.00\ \Omega$ exactly, with the same leads.

Exercise 5.1 When do you need 4 wires?

Same leads ($R_c = 0.1\ \Omega$ each). (a) What is the 2-wire error for $R_x = 100\ \Omega$? (b) Below which R_x does the 2-wire error exceed 1%? (c) The sense leads have resistance too — why doesn't it matter?

Solution

(a) $R_{meas} = R_x + 2R_c = 100.2\ \Omega$: error $0.2/100 = 0.2\%$ — 2-wire is fine for large R_x . (b) Error $= 2R_c/R_x > 1\% \Leftrightarrow R_x < 200R_c = 20\ \Omega$. (c) The sense inputs go to a high-impedance amplifier, so the sense current is negligible: no current through the sense-lead resistance means no voltage drop across it.

5.2 AC measurements (RMS)

For AC the meter reports the RMS value. A *true-RMS* meter integrates v^2 directly; cheaper meters measure the peak or rectified average and *assume a sine*, applying a fixed multiplier — so they read wrongly for non-sinusoids. For a sine, $V_{RMS} = 0.707 V_p = 1.11 V_{avg}$. If the signal has a DC component, $V_{RMS} = \sqrt{V_{RMS,AC}^2 + V_{DC}^2}$.

Protocol 5.2 Get the true RMS from a meter reading

1. Identify the waveform and what the meter senses (true-RMS, peak, or average).
2. For a sine: $V_{RMS} = V_p/\sqrt{2}$; $V_{avg} = 2V_p/\pi$; $V_{RMS} = \frac{\pi}{2\sqrt{2}} V_{avg} \approx 1.11 V_{avg}$.
3. Non-zero average: combine quadratically, $V_{RMS} = \sqrt{V_{RMS,AC}^2 + V_{DC}^2}$.
4. For a non-sine on a sine-calibrated meter, undo its multiplier and apply the correct one.

Worked Example 5.2 Sine plus offset

$v(t) = 1.5 \sin \omega t + 2$ (V): an all-positive sine, offset 2 V. The AC part is $V_{RMS,AC} = 1.5/\sqrt{2} = 1.06$ V and $V_{DC} = 2$ V, so $V_{RMS} = \sqrt{1.06^2 + 2^2} = \sqrt{5.125} = 2.26$ V. A meter measuring only the AC-coupled part reads 1.06 V; you then add the DC quadratically.

Exercise 5.2

A meter that senses the *average* (sine-calibrated, $\times 1.11$) is given a 10 V peak *square* wave. What does it display, and what is the true RMS?

Solution

For a square wave $V_{avg} = V_p = 10$ V, so the meter shows $1.11 \times 10 = 11.1$ V — wrong. The true RMS of a square wave is $V_{RMS} = V_p = 10$ V. (A true-RMS meter would read 10 V.)

Chapter 6

Laboratory: Bench measurements

This lab puts Chapters 3–5 to work on a real bench: a Rigol DS1054 digital storage oscilloscope (DSO) observing signals generated by a signal-generator board. The point is not the individual numbers but the *method* — calibrate, choose scales, read both with the instrument’s engine and by eye, and attach an uncertainty to everything.

Key Idea

Every reading here is taken *twice*: once from the DSO’s automatic measurement engine (uncertainty from the datasheet) and once by *visual* inspection in divisions (uncertainty ≈ 0.1 div). Agreement between the two is the cross-check that the measurement is sound.

6.1 Setup and probe calibration

What is happening: a $\times 10$ probe forms an RC network with the scope input, so it must be *compensated* or fast edges are distorted (Appendix B.1). The scope provides a 1 kHz, 3 V square reference for exactly this.

Protocol 6.1 Calibrate the bench before measuring

1. Connect each probe to the PROBE COMP terminal; select a timebase around 200 $\mu\text{s}/\text{div}$.
2. Turn the trimmer on the BNC until the square wave has flat tops — no overshoot (over-compensated) and no rounded droop (under-compensated).
3. Switch the display to *dots* (single samples, minimum persistence) so individual acquisition points are visible — needed for the next section.

6.2 Measuring the sampling rate

A budget DSO has *one* ADC time-division-multiplexed across the active channels. With the display in dots mode you count how many samples fall across a known time window and back out the sample period

$$T_s = \frac{(\text{timebase}) \times (\# \text{ divisions})}{\# \text{ dots}}, \quad \Delta T_s = \frac{T_s}{\# \text{ dots}},$$

and the sample rate $f_s = 1/T_s$ carries the *same relative* uncertainty.

Worked Example 6.1 The sample rate halves with each channel

On CH1 alone, 60 dots span 12 divisions at 5 ns/div: $T_s = (5 \times 12)/60 = 1$ ns, i.e. $f_s = 1$ GSa/s. Enabling CH2 drops the count to 30 dots ($f_s = 500$ MSa/s); a third channel

halves it again to 250 MSa/s. One ADC, shared in turns — the rate divides as channels are added.

6.3 Analog waveform characterisation

For each generated waveform (sine, triangle, square) the protocol is the same.

Protocol 6.2 Characterise a periodic waveform

1. Trigger a stable trace; set vertical and horizontal scales so the wave fills most of the screen (best resolution, §1.4).
2. Read V_{pp} , V_{min} , V_{max} , frequency, period, average and V_{RMS} from the engine; record the datasheet uncertainty for each.
3. Re-read the same quantities *by eye* in divisions $\times$ scale; the reading uncertainty is about 0.1 div.
4. For the square wave additionally read rise and fall times (t_{rise} , t_{fall}) on a fast timebase (Appendix C).
5. Compare engine vs visual: they should agree within their uncertainties.

A representative engine reading for the sinusoid:

Table 6.1: Sinusoid, DSO engine (datasheet uncertainty).

Quantity	Value	Δ	ε
V_{pp}	1.14 V	0.048 V	4.2 %
V_{max}	1.58 V	0.080 V	5.6 %
Frequency	7999 Hz	0.20 Hz	0.0025 %
V_{RMS}	1.07 V	0.080 V	7.5 %

Note

The frequency is pinned down far better ($\varepsilon \sim 10^{-3}$ %, from the timebase) than the voltages ($\varepsilon \sim$ several %, from the gain accuracy). Time is the scope's strong suit; amplitude is not.

6.4 Digital signal characterisation

For the digital outputs (high- and low-frequency, switches 4 and 3) the extra quantities of interest are the *duty cycle* (Duty+/Duty−) and the edge times. A high-frequency square near 2.35 MHz shows a short $t_{rise} \approx t_{fall} \approx 20$ ns; the low-frequency one (~ 24 Hz, $\approx 50\%$ duty) shows much longer periods but comparable edges.

Watch Out

At high frequency the probe and scope bandwidth limit the edges you can *see*: a measured t_{rise} near the scope's own rise time ($t_{rise}^{scope} = 0.35/BW$, Appendix C) is the *instrument*, not the signal. Always check the edge against the scope's bandwidth before trusting it.

Appendix A

Uncertainty propagation

For $y = f(x_1, \dots, x_n)$ a first-order Taylor expansion about the measured point gives $\Delta y = \sum_i |\partial f / \partial x_i| \Delta x_i$. Hence: sum/difference $y = a \pm b \Rightarrow \Delta y = \Delta a + \Delta b$ (absolute add); product/ratio $y = ab$ or $a/b \Rightarrow \varepsilon_y = \varepsilon_a + \varepsilon_b$ (relative add). A difference of nearly equal numbers has a small y_0 but a Δy of the same size as for a sum, so $\varepsilon_y = \Delta y / y_0$ becomes large — the cancellation seen in Chapter 2.

Appendix B

Oscilloscope formulas

Deflection: $x_{ndiv} = V_x/K_x$, $y_{ndiv} = V_y/K_y$. Time base: $A = K_x N_{xdiv}/2$, $K_T = T/N_{xdiv}$. Scale choice: vertical $K_y \geq (V_{max} - V_{min})/8$, horizontal $K_x \geq NT/12$ (1–2–5 sequence). Reading: $V = KL$, $\varepsilon_V = \varepsilon_K + \varepsilon_L$.

B.1 The compensated probe

With $C_p = C_i + C_c$ the uncompensated input has a pole at $f_t = 1/(2\pi R_g C_p)$. Adding a probe $R_s \parallel C_s$ with $R_s = 9R_i$, $C_s = C_p/9$ (so $R_s C_s = R_i C_p$) makes a frequency-independent divider of 1/10 and pushes the pole to $10f_t$ — ten times the bandwidth, at the cost of a $\times 10$ attenuation.

Appendix C

Bandwidth and rise time

A single-pole bandwidth B responds to a step as $v(t) = V_H(1 - e^{-t/\tau})$ with $\tau = 1/(2\pi B)$; the 10%–90% rise time is $T_B = \tau \ln 9 = 0.35/B$. A real edge T_{Rise} and the scope combine as $T_{meas}^2 = T_B^2 + T_{Rise}^2$, so $T_{Rise} = \sqrt{T_{meas}^2 - T_B^2}$; near $T_{meas} \approx T_B$ this subtraction is ill-conditioned (large $\varepsilon_{T_{Rise}}$).

Appendix D

RMS, average and peak factors

Waveform	V_{RMS}	V_{avg} (rectified)	note
Sine $V_p \sin \omega t$	$V_p/\sqrt{2} = 0.707V_p$	$2V_p/\pi = 0.637V_p$	$V_{RMS} = 1.11V_{avg}$
Square $\pm V_p$	V_p	V_p	sine meter reads $1.11V_p$
With DC	$V_{RMS} = \sqrt{V_{RMS,AC}^2 + V_{DC}^2}$		

Appendix E

Example coverage map

Example(s)	Protocol / method	Where
SI redefinition; 7 base units; traceability; direct vs indirect	Take & trace a measurement (P 1.1)	Ch. 1
Sum/diff/product/ratio; ratio-2 ($1/(a-b)$); R_{in} divider; powers (V^2/R)	Propagate uncertainty (P 2.1)	Ch. 2
Scope Examples 1–10 (spots, waves, time base)	Place spot / read trace (P 3.1)	Ch. 3
Vertical/horizontal scale choice	Pick scales (P 3.2)	Ch. 3
V_{pp} uncertainty (K_y, L); period/frequency reading	Reading uncertainty (P 3.3)	Ch. 3
Sampling, Nyquist, interpolators, aliasing; alias-frequency prediction	Avoid aliasing (P 4.1)	Ch. 4
Compensated probe; rise time Ex1/Ex2	Rise time (P 4.2)	Ch. 4
DMM loading; 2-/4-wire resistance	2-/4-wire (P 5.1)	Ch. 5
AC RMS, sine/square, $1.5 \sin + 2$	True RMS (P 5.2)	Ch. 5
Probe compensation; sampling rate; analog/digital waveform reads	Calibrate / characterise (P 6.1–6.2)	Ch. 6